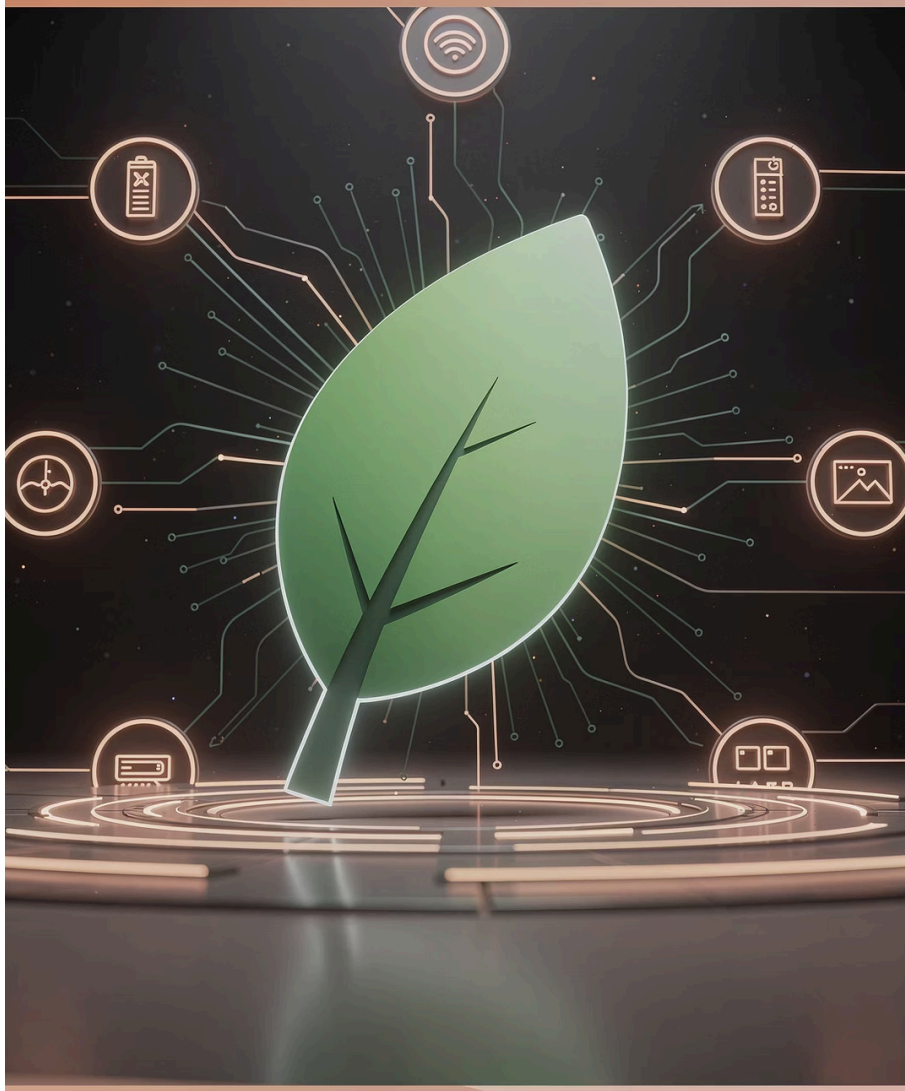
MongoDB

Getting started with the world's leading NoSQL database



Installation Setup

Install **2 things** to get started:

MongoDB Community Server

Download here

MongoDB Shell

Download here

🪟 Windows

Full guide: YouTube Tutorial

🍎 Mac

Full guide: YouTube Tutorial

Video shows v6.0 — install **v7.0** (latest).
Commands remain the same.

mongosh

The modern MongoDB Shell for interacting with your database

Start mongosh

Just type `mongosh` in your terminal.

`show dbs`
List all databases

`use <db>`
Switch to or create a database

`show collections`
List collections in the current database

`db.createCollection("name")`
Create a collection

`db.<collection>.insertOne({})`
Insert a document

`db.<collection>.find()`
Query all documents

`db.<collection>.updateOne()`
Update a document

`db.<collection>.deleteOne()`
Delete a document

`exit`
Exit the shell

mongosh & JavaScript

mongosh is a full JavaScript interpreter — you can run JS directly in the shell.

`Math.round(2.6)`

Use built-in JS math functions

`let x = 10; x * 2`

Declare variables and evaluate expressions

`new Date()`

Use JS Date objects

`[1,2,3].forEach(n => print(n))`

Use array methods and arrow functions

`function greet(name) { return "Hello, " + name }`

Define and call functions

`db.users.find().forEach(doc => print(doc.name))`

Combine JS with MongoDB queries

Any valid JavaScript expression can be typed directly into the mongosh prompt.

JSON vs JavaScript Object

They look almost identical — but have key differences.

JavaScript Object

```
{
  name: "Alice",
  age: 30,
  active: true
}
```

- Keys don't need quotes
- Values can be functions
- Only exists in memory (runtime)

JSON

```
{
  "name": "Alice",
  "age": 30,
  "active": true
}
```

- Keys MUST be in double quotes
- No functions allowed
- Used for storing & transferring data (text format)

`JSON.stringify(obj)`

JS Object → JSON string

For sending/storing

`JSON.parse(str)`

JSON string → JS Object

For using in code

Think of JSON as a 'shipping format' — you convert to JSON to send data, and convert back to use it.

BSON Data Types

MongoDB stores data as BSON (Binary JSON)

String
`"hello"` — UTF-8 text

Integer
`42` — 32-bit or 64-bit number

Double
`3.14` — Floating point number

Boolean
`true` / `false` — Boolean value

Date
`new Date()` — Date and time (stored as ms since epoch)

ObjectId
`new ObjectId()` — Unique 12-byte document identifier

Array
`[1, 2, 3]` — List of values

Object
`{ key: value }` — Embedded/nested document

Null
`null` — Null value

Binary
`BinData(0, "...")` — Binary data (e.g. images, files)

Every document automatically gets an `_id` field of type ObjectId if not provided.

BSON vs JSON — Why BSON Wins

BSON is MongoDB's storage format. Here's why it's better than plain JSON for a database:

More Data Types

JSON only has: string, number, boolean, array, object, null

BSON adds: Date, ObjectId, Binary, Int32, Int64, Decimal128, and more

Dates as Real Values

JSON stores dates as strings → `"2024-01-01"` (just text, no math)

BSON stores dates as actual timestamps → `new Date()` (sortable, comparable)

Efficient Binary Storage

BSON is binary (not text) → faster to read/write for a database engine

JSON is plain text → slower to parse at scale

Precise Numbers

JSON has one "number" type — no distinction between int and float

BSON has Int32, Int64, Double, Decimal128 → no precision loss

Unique IDs Built-in

BSON has `ObjectId` — a compact, auto-generated unique ID

JSON has no native ID type

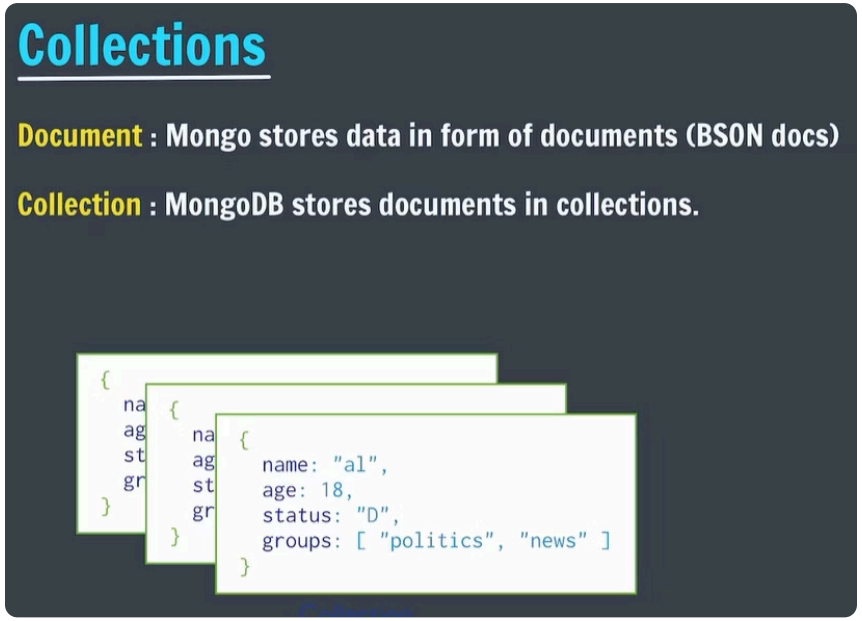
Supports Binary Data

BSON can store raw binary (images, files) directly

JSON cannot — you'd need to encode it as Base64 string first

BSON is designed for machines. JSON is designed for humans. MongoDB uses BSON internally but lets you work with JSON-like syntax.

Collections & Documents



If you know SQL, MongoDB maps directly to concepts you already know.

SQL	MongoDB	Description
Database	Database	The top-level container
Table	Collection	A group of related data
Row / Record	Document	A single entry of data
Column	Field	A single property/attribute
Primary Key	`_id`	Unique identifier per record
JOIN	Embedded Doc / `$lookup`	Linking related data

SQL Table: users

```
| id | name  | age | status |
|----|-------|-----|--------|
| 1  | Alice | 25  | active |
| 2  | Bob   | 30  | inactive |
```

MongoDB Collection: users

```
{ _id: ObjectId(), name: "Alice", age: 25, status: "active" }
{ _id: ObjectId(), name: "Bob",   age: 30, status: "inactive" }
```

Key difference: MongoDB documents in the same collection don't need to have the same fields — they're flexible, unlike SQL rows which must match the table schema.

INSERT in DB

insertOne

`db.collection.insertOne()`	Inserts a single document into a collection.

show collections

db.student.insertOne({ name: “adam”, marks: 79 })

db.student.find( )

If a collection does not exist, MongoDB creates the collection when you first store data for that collection.

```
test> use college
switched to db college
college> show collections

college> db.student.insertOne( {name: "adam", age: 19, marks: 88} )
{
  acknowledged: true,
  insertedId: ObjectId("64fec88c4d37352da5df8407")
}
college> db.student.find()
[
  {
    _id: ObjectId("64fec88c4d37352da5df8407"),
    name: 'adam',
    age: 19,
    marks: 88
  }
]
college>
```

INSERT in DB

insertMany (array of documents)

db.student.insertMany([{ name: “bob”, marks: 65 }, { name: “eve”, city: “Delhi“ }])

`db.collection.insertMany()`	Inserts multiple documents into a collection.

FIND in DB

db.collection.find( ) //returns everything

for specific queries

db.collection.find({ key: value })

db.collection.findOne({ key: value })

```
college> db.student.find( {city: "Delhi"} )
[
  {
    _id: ObjectId("64fec9714d37352da5df8408"),
    name: 'bob',
    city: 'Delhi',
    marks: 75
  },
  {
    _id: ObjectId("64fec9f54d37352da5df8409"),
    name: 'catlyn',
    marks: 64,
    city: 'Delhi'
  }
]
```

FIND in DB

Query Operators

Q. Find students where marks > 75

```
db.student.find( {marks: {$gt: 75}} )
```

Q. Find students who live in Delhi or Mumbai

```
db.student.find( {city: {$in: ["Delhi", "Mumbai"]}} )
```

Q. Find students who scored > 75 or live in Delhi

```
db.student.find( {$or: [ {marks: {$gt: 75}}, {city: "Delhi"}]} )
```

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mongodb

Query Predicates – Database Manual – MongoDB Docs

Learn about query predicates in MongoDB. Use these expressions to filter documents in ad-hoc CRUD commands and aggregation pipelines

```
college> db.student.find({city: "Delhi", marks: 75})
[
  {
    _id: ObjectId("64fec9714d37352da5df8408"),
    name: 'bob',
    city: 'Delhi',
    marks: 75
  }
]
college> db.student.find( {marks: {$gt: 75}} )
```

Update

Updates the **first matching document** in a collection.

```
db.collection.updateOne(
  { filter },      // which document to find
  { $set: { ... } } // what to change
)
```

db.collection
The collection you want to update (e.g. `db.users`)

{ filter }
The condition to find the document (e.g. `{ name: "Alice" }`)

$set: { ... }
The fields to update (e.g. `{ age: 26 }`)

Only one document
Even if multiple match the filter

Real example

```
db.users.updateOne(
  { name: "Alice" },
  { $set: { age: 26, status: "active" } }
)
```

Result

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
```

$set
Set/update a field value

$unset
Remove a field

$inc
Increment a number field

$push
Add item to an array

$pull
Remove item from an array

Use `updateMany()` if you want to update ALL matching documents instead of just the first one.

UPDATE in DB

Update Operators

- `$addFields`
- `$set`
- `$project`
- `$unset`
- `$replaceRoot`
- `$replaceWith`

`db.collection.updateMany()`	Update all documents that match a specified filter.
`db.collection.replaceOne()`	Replaces at most a single document that match a specified filter even though multiple documents may match the specified filter.

Nesting

```
{
  _id: ObjectId("64fdfcad780181c3d03ec623"),
  name: 'farah',
  performance: { marks: 88, grade: 'A' }
}
```

to find

```
db.student.findOne( {"performance.marks": 88} )
```

DELETE in DB

deleteOne

db.collection.deleteOne(<filter> , <options>)

deleteMany

db.collection.deleteMany(<filter> , <options>)

db.dropDatabase( )

```
college> db.student.deleteOne( {state: "Haryana"} )
{ acknowledged: true, deletedCount: 1 }
college> db.student.find()
```